

BINOMIAL PROBABILITIES

08/12/25

1. A fair coin is flipped 20 times. What is the probability of exactly ten heads and ten tails are flipped (in any order)?
2. Ten fair dice are rolled. What is the probability that exactly four of them show a square number?
3. Approximately 10% of the population are left-handed. In a class of 14 pupils, what is the probability that exactly 2 of them are left-handed?
4. Due to a manufacturing defect, 12% of a stock of ball point pens are defective. If you buy a pack of 50 pens from this batch at a clearance sale, then what is the probability that exactly 6 of them will be defective?
5. A fifth former arrives late for his maths lesson once a week, on average. Given that the fifth form have four lessons a week, what is the probability that over the next three weeks he is late
 - a. exactly three times;
 - b. exactly once in each of the three weeks;
 - c. exactly three times in exactly one of the weeks (and on time otherwise)?
6. A particular game is won if, on the roll of two dice, you throw either double 6 or no 6s at all. When ten people play this game as a group, the group is declared victorious if no one wins the game, or if exactly 6 people win the game. What is the probability that a group of ten is victorious?
7. Colour blindness affects approximately 1 in 12 men, and 1 in 200 women. In a group of 10 men and 20 women, what is the probability that exactly two people are colour blind.
8. If you are dealt a five-card hand from a normal 52-card deck, then what is the probability that you have exactly three cards from the same suit?
9. If you are dealt a six-card hand from a normal 52-card deck, then what is the probability that you have exactly three cards from the same suit?
10. When Dr Simons tries to throw a pen into a bin at distance there is a 0.2 probability that he gets the pen in. If he misses, then the probability that he gets the next pen in remains 0.2. Once he gets a pen in the bin, however, the probability that he gets any subsequent pen in is 0.9. If Dr Simons throws 10 pens at the bin (one-by-one), find the probability that he gets exactly 4 pens in the bin.
11. A darts training game involves the player throwing three darts at 18, three darts at 19 and three darts at 20. The final score is the total number of accurate hits. I model the probability that I successfully hit 18 as 0.4; that I successfully hit 19 as 0.2; and that I successfully hit 20 as 0.6. Find the probability that I score exactly 3 when playing this game.
12. My bookcase contains 6 maths books, 4 physics books, and two books of Jokes. Every morning I choose a book at random from the bookcase, read a chapter, and then replace the book. For a week (of seven days), find the probability that:
 - a. I read a joke book at least once;
 - b. I read a physics book at least twice;
 - c. I read a maths book three times, a physics book twice, and a joke book twice;
13. A bag contains 2 red balls, 3 blue balls, and 5 white balls. Ten balls are sampled from the bag with replacement. Find the probability that:
 - a. there are 5 white balls and 5 blue balls (and no red balls);
 - b. there are 2 red balls, 3 blue balls, and 5 white balls;
 - c. there are 3 red balls, 4 blue balls, and 3 white balls;
 - d. exactly 4 of the balls are white;
 - e. all the balls are blue and white;
 - f. at least 5 of the balls are red or blue;
 - g. there are at least 3 red balls, given that there are exactly 5 white balls.

14. Should you sample with or without replacement?

You have two identically-looking urns, each containing red and black balls, and nothing else. One urn contains two red balls and one black ball, and the other contains 101 red balls and 100 black balls. An urn is chosen at random by your friend, and you win a prize if you correctly name the urn on the basis of the evidence of two balls drawn from it. After the first ball is drawn, and its colour noted, you can decide whether or not this ball should be replaced in the urn before the second ball is drawn.

What is your best strategy? And what is the probability that you will win the prize if you follow this strategy?